

ATIKOKAN, ON, Canada

WMO: 717470

Lat: **48.7610N**Lon: **91.6283W**Elev: **1277**StdP: **14.03**Time Zone: **-5.00 (W05)**Period: **94-19**WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.5	-22.9	-35.7	0.8	-28.4	-29.7	1.1	-22.7	16.1	3.0	14.0	6.0	1.4	250	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	84.3	68.6	81.3	66.4	78.5	64.6	71.5	80.4	69.4	77.7	67.4	74.7	7.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.5	109.8	76.2	66.4	102.1	73.8	64.6	95.7	71.7	36.2	80.6	34.3	78.0	32.7	74.9	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.4	11.1	-34.7	89.5	7.2	2.9	-39.9	91.6	-44.1	93.3	-48.1	95.0	-53.3	97.1		
(5)			-34.6	74.8	7.1	2.7	-39.7	76.7	-43.8	78.3	-47.8	79.8	-53.0	81.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.4	4.2	8.8	22.6	37.0	50.4	60.7	65.3	63.1	55.1	41.3	26.3	11.6
	DBStd	23.71	14.82	14.15	13.75	9.90	8.63	7.01	5.63	6.00	8.22	8.56	11.03	13.88
	HDD50	6172	1419	1154	852	402	101	6	0	1	39	298	711	1189
	HDD65	10272	1884	1574	1316	841	459	162	66	109	311	735	1161	1654
	CDD50	1555	0	0	1	11	113	328	474	407	192	28	1	0
	CDD65	181	0	0	0	0	7	34	75	51	14	0	0	0
	CDH74	1897	0	0	0	4	115	403	736	493	141	5	0	0
	CDH80	395	0	0	0	0	23	86	164	102	20	0	0	0
Wind	WSAvg	4.6	4.4	4.6	4.9	5.2	5.2	4.5	4.1	4.1	4.3	4.9	5.0	4.4
Precipitation	PrecAvg	31.4	1.4	1.0	1.6	2.0	3.0	4.3	4.2	4.0	3.6	3.1	1.9	1.5
	PrecMax	39.7	5.2	1.9	3.7	4.6	6.0	6.7	6.7	10.6	9.8	7.4	4.0	3.1
	PrecMin	24.0	0.4	0.3	0.2	0.4	0.7	2.1	1.1	1.0	1.1	0.9	0.6	0.7
	PrecStd	4.7	1.0	0.4	0.9	1.2	1.4	1.3	1.5	2.1	2.1	1.7	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.8	44.9	61.2	71.0	83.3	87.8	88.3	86.9	82.5	72.6	57.2	41.0
		MCWB	33.2	37.8	50.8	52.9	64.2	71.9	70.7	70.0	67.2	59.5	49.5	36.2
	2%	DB	33.3	38.3	51.8	64.5	77.1	81.9	84.1	82.8	78.1	66.1	51.0	36.0
		MCWB	31.6	34.0	43.3	48.6	59.3	65.7	68.7	68.4	65.4	56.0	46.1	33.2
	5%	DB	30.1	34.3	46.0	59.7	72.1	78.5	81.2	79.4	74.0	60.7	45.3	33.0
		MCWB	28.4	31.6	39.2	46.1	56.4	63.4	67.1	66.3	63.2	53.0	40.9	31.5
	10%	DB	25.9	30.3	40.9	54.8	67.5	75.3	78.1	76.1	69.9	55.3	41.0	30.8
		MCWB	24.6	28.0	35.2	43.3	54.3	61.9	65.2	64.5	61.2	49.3	37.9	29.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.4	39.5	51.8	56.3	67.2	72.7	73.9	73.6	70.7	62.5	51.8	36.8
		MCDB	36.7	44.0	57.4	66.4	77.3	84.2	83.4	82.0	78.6	68.1	55.0	39.5
	2%	WB	32.1	34.8	44.6	50.8	62.9	69.1	71.7	70.7	67.7	57.6	45.9	33.6
		MCDB	33.7	37.6	50.9	62.0	71.6	78.5	80.2	79.6	74.8	65.1	51.0	35.2
	5%	WB	28.5	31.9	39.5	47.2	59.2	66.4	69.6	68.5	65.0	53.8	41.3	31.8
		MCDB	30.2	34.7	44.9	57.9	67.6	73.8	77.7	76.1	71.8	59.9	44.8	33.1
	10%	WB	24.8	27.9	35.7	44.1	56.0	64.1	67.5	66.4	62.3	50.1	37.8	29.3
		MCDB	26.1	30.3	40.6	54.4	65.5	71.6	74.7	73.6	68.7	54.4	41.0	30.7
Mean Daily Temperature Range	5% DB	MDBR	19.8	22.9	23.1	22.3	22.5	21.2	21.1	21.2	19.9	15.8	13.4	16.4
		MCDBR	17.8	22.7	25.1	31.3	30.5	26.3	25.1	25.2	25.3	25.0	20.4	16.2
		MCWBR	15.6	18.0	16.8	18.2	15.5	12.5	12.1	12.6	14.3	15.8	14.7	13.7
	5% WB	MCDBR	16.5	18.7	23.5	27.7	23.9	20.8	21.3	21.4	21.5	22.9	18.4	15.1
		MCWBR	14.8	15.2	16.6	17.2	13.7	11.9	11.7	12.0	13.2	15.4	13.9	13.3
Clear-Sky Solar Irradiance	taub		0.254	0.262	0.295	0.329	0.349	0.370	0.383	0.364	0.354	0.306	0.285	0.254
	taud		2.371	2.351	2.404	2.447	2.446	2.406	2.395	2.478	2.479	2.591	2.525	2.431
	Ebn at Noon		263	287	294	293	289	282	277	278	270	270	251	249
	Edh at Noon		21	27	30	33	34	36	36	31	29	22	18	17
All-Sky Solar Radiation	RadAvg		388	766	1174	1522	1673	1776	1832	1564	1132	662	354	280
	RadStd		57	84	101	148	98	143	108	122	91	90	40	29